

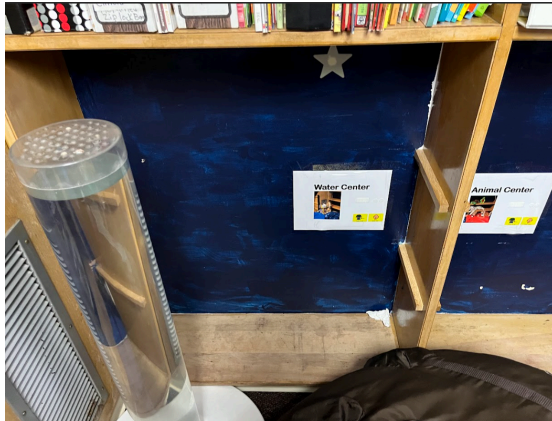
Element F: Consideration of Design Viability

Before we can start the manufacturing process, it is important to assess risks and potential failures. When considering potential failures, we limited them to our infrastructure design phases 1 and 2, which includes covers for the middle island and covers for the cubbies. Because we are planning on pushing off the phase 3 tactile panels, they were not considered. We have come up with the following list of potential shortcomings and their proposed solutions.

Problems and Solutions:

Problems	Solutions
There are potential issues with our current pocket door design and its translation to the physical area currently suiting the middle island and cubbies in the sensory area. 	- We may need to make changes to existing infrastructure - This includes removing shelving, changing position and or shape of posts/supports, etc. - We can remove and shorten the shelves, and move the shelf supports to the back wall of the island so that there is no interference with the pocket door design. - We could potentially replace the supports with triangle supports, or supports that are supported by vertical panels.

- Designed sliding panel cover not ‘fitting’ in cubbies and their current infrastructures



- We may need to make changes to existing infrastructure
- We may have to remove and change the position of shelving supports if they interfere with the sliding panel design.
- We may have to attach the casing for the sliding cover to the existing border of the cubbies.
- This would allow the cover to slide in and be seated on the base of the casing, rather than sliding to the floor below the cubby, or cutting into the shelf above.

- We originally planned on covering 4 cubbies, but after talking with stakeholders, that was shortened to 2 or 3 cubbies as one could not be touched without custodial/authority approval.
- There is one cubby in question pictured below, that serves as both a tactile wall and a seating area for kids interacting with the bubble tube, which cannot be moved.

- Ideally, we will have enough storage with the cubbies we can work with.
- We have decided to not use the cubbies in the back as the kids seem to be using the area enough that it would be a major inconvenience to change.
- We will use some kind of shelving or stacked container design inside the middle island and cubbies in order to optimize the space for storage.
- If necessary, we can apply our cubby cover design to a ‘cubby’ on the left side of the back wall that is currently storing items, but not at full capacity.
 - This ‘cubby’ having a cover would add some consistency to the theme of the overall sensory room.



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- Timing of the project could be an issue because we need to make sure that we deliver what we have promised to the customer on time.
- The installation of the middle island covers and cubby covers will be challenging when the kids are always in the classroom the entire day.

- The doors on the middle island will only be able to hold up to 30 pounds so leaving obvious points for the kids to pull at the door could be a problem.

- To make sure that we deliver the top priority of the customer being the middle island cover, we splitted the project into 2 phases, one being the middle island, and second being the side cubbies.
- Splitting the project into 2 phases also allows us to install the covers individually so that we don't interrupt the kid's time in the sensory room.

- We can put a panel on the doors that has the cut outs for the tactile device while not reducing the structure of the doors.

To further assess the viability of our project before manufacturing, we compiled a bill of materials, totaling the costs associated with the infrastructure phases of our project.

Bill of Materials:

<u>Item and Description</u>	<u>Cost</u>
Pocket door	\$57.99*4= \$231.96
Actual pocket hinge	\$31.99*4= \$127.96
Bi-Fold Door Hinges Online (x3)	\$4.96*3=\$14.88
Purebond Birch Plywood	\$79
Cubby wood	\$26*2= \$52
Wood Stain	\$19.98
Total value	≈\$525.44

When considering total cost, phase 1 would not include cubby wood and the wood stain is provided for by the school. This means that our total spending will be less than the total value listed above. We discussed the costs and our manufacturing plan with our supervisor/teacher and he deemed the cost suitable for our project, especially considering that our project focuses on infrastructure that cleans up the sensory area as a whole and provides a platform for future sensory related projects in the sensory room. Room will be freed on the back and side walls because of the storage room provided by the middle island, and eventually side cubbies.